

SYSTEMS AND METHODS FOR GENERATING A USER-SPECIFIC ANATOMICAL GEOMETRY PROFILE VIA TIME-OF-FLIGHT RANGING AND OPTICAL MOTION SENSING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. _63/983,946_, filed _02/16/2026 03:21:42 PM Z ET_, the contents of which are incorporated herein by reference in their entirety. In certain embodiments, the systems and methods described herein may be implemented independently or as part of a multi-node biometric data platform comprising additional physiological sensing systems, structural mapping architectures, comparative modeling engines, calibration frameworks, machine learning synthesis modules, or user-specific personalization infrastructures. The present disclosure is directed specifically to geometry acquisition and profile generation architectures and remains operable as a standalone system independent of such integration.

FIELD OF THE INVENTION

The present disclosure relates generally to biometric measurement systems and anatomical profiling technologies. More specifically, the disclosure relates to systems and methods for generating a user-specific penile anatomical geometry profile using time-of-flight (ToF) distance ranging, optical motion sensing, inertial measurement stabilization, and computational surface reconstruction techniques.

In various embodiments, the disclosure pertains to portable, handheld, body-positioned, or semi-fixed devices configured to capture spatial measurements of penile anatomy under standardized measurement conditions and controlled orientation parameters. The systems described herein may incorporate one or more ToF ranging modules configured to measure radial distance between a sensing reference plane and a target anatomical surface, in combination with optical motion sensors configured to detect longitudinal displacement relative to the anatomical structure. In certain embodiments, inertial measurement units (IMUs) may be used to compensate for angular deviation, tilt, or unintended user movement during scanning.

The measured distance and motion data may be processed through computational reconstruction algorithms configured to generate a structured three-dimensional geometric representation of the scanned anatomy. The resulting user-specific anatomical geometry profile may include derived structural metrics including but not limited to total longitudinal length, localized cross-sectional radius distribution, circumferential variability, curvature mapping, volumetric estimation, proportionality indices, taper gradients, symmetry metrics, surface continuity parameters, and parametric geometric descriptors. Such geometry profiles may be stored digitally, compared over time, utilized for device calibration, integrated into compatibility modeling systems, or analyzed within broader biometric analytics platforms.